



Indian Institute of Technology Kharagpur

Date of Examination:

Duration: 1 Hour

Subject No.:

Subject:

Department/Center/School: Computer Science & Engg. Credits: 4 Full marks: 15

Name: _____

Roll Number: _____

Instructions

- i. This question paper contains 8 pages and 2 questions. All questions are compulsory. Marks are indicated in parentheses. This question paper has been cross checked.
- ii. Please write your name, roll number, subject name and code, date and time of examination on the answer script before attempting any solution.
- iii. Use of electronic calculators only is permitted. No extra resources viz. graph papers, log-tables, trigonometric tables would be required.
- iv. **Organize your work**, in a reasonably neat and coherent way. Work scattered all across the answer script without a clear ordering will receive very little marks.
- v. **Mysterious or unsupported answers will not receive full marks.** A correct answer, unsupported by calculations, explanation, will receive no marks; an incorrect answer supported by substantially correct calculations and explanations may receive partial marks.
- vi. Write your answers in the space provided for that question. If the space provided for the answer is not sufficient, the rest of the answer may be written in the extra space provided after all questions. This should be clearly indicated in the space provided, failing which, the extra answer may not be checked.

Question:	1	2	Total
Points:	7	8	15
Score:			

1. Let $x_1, x_2, \dots, x_n$ be independent and identically distributed (iid) random variables drawn from a distribution with probability density function

$$f(x) = \begin{cases} \alpha x^{-(\alpha+1)}, & x \geq 1, \\ 0, & x < 1, \end{cases}$$

where $\alpha > 0$ is an unknown parameter. Assume that the observed data $x_1, x_2, \dots, x_n$ satisfy $x_i \geq 1$ for all i .

- (a) (3 points) Show that the *maximum likelihood estimator (MLE)* of α is, $\hat{\alpha}_{\text{MLE}} = \frac{n}{\sum_{i=1}^n \log x_i}$.

- (b) (4 points) Suppose the prior distribution of α is a *Gamma distribution* with density

$$p(\alpha) = \frac{b^a}{\Gamma(a)} \alpha^{a-1} e^{-b\alpha}, \quad \alpha > 0,$$

where $a > 1$ and $b > 0$ are known constants. Show that the *maximum a posteriori (MAP)* estimator of α is, $\hat{\alpha}_{\text{MAP}} = \frac{n+a-1}{b + \sum_{i=1}^n \log x_i}$.

2. (a) (3 points) Let X be a positive continuous random variable and let the scalar value $a > 0$. prove the following relation. [Hint: Start with the expression of expectation. Then use the property that X is a positive random variable to modify integration limits. Next split the resulting integration so that you can ultimately get $P(X \geq a)$ from it.]

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

- (b) (5 points) Let X be a random variable following a uniform distribution $\mathcal{U}[0, \theta]$. In other words,

$$p(X = x) = \begin{cases} 1/\theta, & 0 \leq x \leq \theta \\ 0, & \text{otherwise} \end{cases}$$

You are given N IID observations $\mathbf{x} = (x_1, x_2, \dots, x_N)^T$ drawn from the distribution above. Find the Maximum Likelihood Estimator (MLE) for θ . (Hint: Write down the likelihood function carefully, paying attention to the range of valid θ values given the observed data.)

The questions end here.

The next few pages are added extra. You can use these, if for any question you need extra space.
